

Zhuocheng (Tom) Qiao

AI Agent Engineer / AI Backend Engineer

Shanghai | M.S., Signal and Information Processing, East China Normal University (2022)

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[Engineering Portfolio](#): Agent Runtime, AI Harness, Open Source & Project Deep Dives



SUMMARY

Microsoft Software Engineer II with 4+ years building production AI agent runtimes, evaluation/agent harnesses, and backend platforms, with hands-on experience in distributed systems, productization, reliability, and cross-team delivery.

EXPERIENCE

Microsoft | Software Engineer II | Jun 2022–Present

Office Agent (Societas) | General-Purpose Agent Platform | Jul 2025–Feb 2026

A general-purpose agent for Microsoft 365 Personal and Family subscribers, reaching **27K WAU**—approximately **8.5%** of eligible users—within **two months** of global launch; led core work across Agent Runtime, productionization, reliability, and compliance engineering.

- **Native Tool Calling**: Led the migration of tool invocation from text-based XML parsing to native function calling, adapting streaming tool-call protocols across OpenAI and Anthropic SDKs and unifying Tool Call/Result execution; designed Native/XML dual-mode rollout and rollback, extended through the full sub-agent chain, improving structured reliability and cross-model consistency.
- **Reliable Message Delivery**: Independently rebuilt long-running agent messaging with Redis Lists for 24-hour replayable state and Service Bus Sessions for retention, acknowledgement, redelivery, and per-session FIFO; introduced a rollout/rollback transport abstraction for backend interruptions and browser reconnect replay.
- **Agent Proxy**: Designed and delivered Agent Proxy end-to-end to production from scratch, implementing an OBO authorization flow for user-delegated access to downstream resources; delivered task persistence, restart recovery, containerization, and CI/CD, enabling reliable external integration of Agent capabilities as a standalone service.
- **Data Security & Compliance**: Designed and implemented a user-level encryption scheme—hierarchical keys, hashed lookup, self-describing ciphertext, rotation, and gradual migration—later reused by partner modules; independently migrated Societas backend node-to-node traffic from HTTP to HTTPS through Key Vault certificate injection, PFX-to-PEM conversion, server certificate loading, and environment-wide cutover, and added CA verification to Agent Proxy downstream traffic, protecting data at rest and in transit.

Microsoft Copilot | Graph-based Agentic Loop & Production Engineering | Mar 2026–Aug 2026

Work spans Epichan's graph-based Harness and Loop Engineering practices, and Studio's production AI product and delivery systems.

- **Epichan (Contributing to Graph/Loop Engineering in Practice)**: Built on Epichan's graph-based Harness (Developer / Reviewer / Judge agents scheduled along a DAG), led the design and iteration of the a11y-remediation-workflow Skill, codifying defect-reproduction flows, role-specific evidence requirements, and independent acceptance methods as reusable Skills; validated ARIA semantics, the accessibility tree, keyboard and focus behavior, and cross-scenario regressions by issue type, then fed recurring failures back into the Skill and regression tests to form a continuous execution-evidence, rule-iteration, and regression-validation loop.
- **Studio | AI Coding Engineering Efficiency & Production Delivery**: For high-frequency AI Coding submissions, optimized the delivery pipeline to increase CI/CD throughput while preserving quality gates; designed and shipped automated flaky-test detection, quarantine, and tracking, and drove change classification and test parallelization, reducing build time by approximately **30%**; developed a working understanding of Copilot's production inference path—from session ingress and Turn/Orchestrator scheduling through model-tool loops, safety/citation processing, and streaming responses—and supported multi-region progressive rollout and production on-call, using release records, health signals, and regional state for incident localization and tiered mitigation.

Office Plus | Commercial Backend & Identity Governance | Jul 2024–Jun 2025

FY25 .NET/ABP commercial platform on Azure: **3.4M new paid subscriptions, \$6.9M revenue**; subscriptions, payments, entitlements, identity, and third-party delivery.

- **Plagiarism checking**: Led upload-to-refund workflows across orders, payments, entitlements, and Service Bus; the feature processed **6,320** orders in its first **10 days**.
- **Subscription and channel growth**: Led WeChat auto-renewal and Partner VIP top-up with decoupled contract/order/entitlement states, layered auth, idempotency, callback isolation, and compensation; after launch, auto-renewal reached **86%** of total orders; business data showed ARPPU +**58%** MoM, and manual renewal rose from **13.8%** to **24%**.
- **Identity and session governance**: Led soft deletion and refresh-token/JWT device-session modernization with recoverable state, audit, and scheduled cleanup; reduced record growth from **MAU × devices × 30** to **MAU × devices** with quotas, revocation, and gradual migration.

Earlier Projects: XPay (Jun–Oct 2022) | PC Manager Client (Nov 2022–Jun 2024)

TECHNICAL SKILLS

Backend/Languages: Python, C#/I.NET, Java, FastAPI, REST/SSE, Distributed Tasks & Asynchronous Processing

Agent Engineering: Function Calling, Multi-Agent/Sub-Agent, Agent Harnesses, Skills, Tool Orchestration, Context & Memory, Automated Evaluation & Quality Gates

Infrastructure: Azure, Service Bus, Redis, PostgreSQL, Docker, CI/CD, Logging & Observability

OPEN-SOURCE CONTRIBUTIONS

Contributed to open-source projects including Hermes Agent and Multica, with **10+ fixes** merged or incorporated upstream across agent security, runtime reliability, tenant isolation, and API robustness; experienced in issue discovery, root-cause analysis, patches, tests, and maintainer collaboration.